Tiam Bennett Morrow-Rogers

Ottawa, Ontario
tiambennett@gmail.com | tiamdesigns.com | github.com/TiamDesigns

Summary

Electrical Engineering student at the University of Ottawa with a completed Bachelor of Industrial Design. Co-founded Phase Studio and took the Titan65 keyboard from prototype to 450+ units sold. Seeking hardware, embedded and validation internships.

Education

University of Ottawa 2024 – Expected 2029
Bachelor of Applied Science, Electrical Engineering *Ottawa, ON*

Humber College 2017 – 2022
Bachelor of Industrial Design; Certificate in Design Foundations *Toronto, ON*

Work Experience

Project Manager September 2025 – June 2026
University of Ottawa MakerLabs *Ottawa, ON*
- Mentored 40+ students across two labs in Arduino, hardware testing, Onshape CAD, 3D printing and laser cutting.
- Reviewed student client projects and gave feedback on concept development, technical execution, testability and lab safety.
- Troubleshot prototyping equipment and helped students debug their circuits and code.

Floor Lead / Computer Technician February 2023 – July 2024
Staples *Cranbrook, BC*
- Diagnosed and repaired customer computer hardware and software issues; trained new staff.

Co-Founder and Product Lead January 2019 – July 2022
Phase Studio *Toronto, ON*
- Took the Titan65 keyboard from prototype to 450+ units sold worldwide; $225,000+ revenue in a single month.
- Avoided board rework on mid-production PCB defects with substitute parts; cost 65% less than the proposed fix.
- Managed DFM, sourcing and QC with suppliers in China; found and prevented a recurring case finish defect.
- Provided technical support to 400+ customers through the company's Discord community.

Freelance 3D Designer April 2022 – July 2022
Automotive Parts Manufacturers' Association *Toronto, ON*
- Modeled 3D product assets in SolidWorks and Fusion 360 for CES 2023 product demonstrations.
- Relayed technical requirements between international engineering teams and the CTO; built UI/UX prototypes in Figma.

Design Researcher November 2021 – May 2022
Humber College *Toronto, ON*
- Built CAD models and functional prototypes to assess manufacturing feasibility; ran cross-faculty design workshops.

Projects

Sample Subway – Nuclear Pressure Tube Sampler 2025
University of Ottawa / Canadian Nuclear Laboratories – Group Lead *tiamdesigns.com/sample-subway*
- Led a four-person multidisciplinary team building a remote sampler for CANDU pressure tubes on a $100 budget.
- Built ESP32-based electronics: HX711 load cell amplifier, 3S Li-ion pack with BMS, buck regulator, H-bridge motor drivers.
- Wrote C++ firmware for load cell readings and motor control, plus a Python host script for Wi-Fi/Bluetooth telemetry.

Dachshund60F – Tri-Mode Wireless Keyboard 2026 – Present
Personal project, AI-assisted *tiamdesigns.com/dachshund60f*
- Designing a USB/Bluetooth LE/2.4 GHz keyboard on the nRF52840: three KiCad PCBs, Zephyr firmware, CNC case.
- Directed Claude Code through schematics, layout and firmware; set the architecture and caught CAD errors it missed.

Technical Skills

Hardware and Lab: KiCad, PCB layout, soldering, oscilloscopes, logic analyzers, SystemVerilog, Vivado, Quartus

Embedded and Programming: C, C++, Python, Arduino, ESP32, nRF52840, Zephyr, MATLAB, Git, JavaScript

CAD and Fabrication: SolidWorks, Fusion 360, Onshape, AutoCAD, Blender, 3D printing, laser cutting, DFM/DFA

Design and UX: Figma, user research, usability testing, KeyShot, Adobe Photoshop, Illustrator, After Effects, Premiere

Awards and Certifications

1st Place, Life Cycle Analysis Competition – uOttawa / Canadian Nuclear Laboratories (2025)

1st Place, Annual Chair Show – Humber College (2020)

Google UX Design Certificate (2022) **Languages:** English (native), Mandarin (intermediate)